Empirical, Behavioral, and Systems Evaluation of Small Language Models Under Severe VRAM Constraints

A Rigorous Systems Engineering and Data Science Investigation on NVIDIA GeForce GTX 1050 Ti

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Abstract

Local on-device inference using Small Language Models (SLMs, 2B–4B parameters) provides verifiable data sovereignty and deterministic latency for autonomous edge nodes and private servers. However, operating under severe physical memory limitations ($\leq$ 4096 MiB VRAM) imposes strict trade-offs across quantization precision, speculative execution, effective memory bandwidth, and safety alignment. This study presents a controlled empirical investigation conducted on an NVIDIA GeForce GTX 1050 Ti GPU (4096 MiB GDDR5, 128-bit memory bus, 112 GB/s theoretical bandwidth, Compute Capability 6.1) hosted in a Proxmox VE LXC container (CT 108). Four candidate models (**Gemma-4-E2B-it**, **Qwen-3.5-4B**, **Qwen-3.5-2B**, and **Llama-3.2-3B**) were benchmarked under the `llama-server` runtime (commit b4120+). Telemetry was captured directly via `libnvidia-ml.so.1` at 250 ms intervals. A controlled Multi-Token Prediction (MTP) speculative decoding ablation with $N = 15$ repetitions isolated a statistically significant **30.9%** overall throughput gain (42.21 ± 7.18 vs 32.25 ± 3.87 tok/s, 95% CI, Welch's $t = 3.139$, $p = 0.0024$, Cohen's $d = 1.146$) and **37.1%** in steady-state warmed execution (46.5 vs 33.9 tok/s), requiring only 2038 MiB VRAM. Controlled quantization of Gemma-4-E2B revealed that IQ4_XS delivers a 59.3% throughput advantage over Q4_K_M in steady-state (41.92 vs 26.34 tok/s, $t = 27.20$, $p = 3.59 \times 10^{-9}$). In safety batteries, all models cleanly refused direct exploits (100% refusal against CVE-2021-44228, stealth keyloggers, and DAN), but universally succumbed to indirect prompt injection embedded in system logs (0% resistance), proving that current alignment mechanisms fail to isolate untrusted context data.

Keywords: Small Language Models, Severe VRAM Constraints, Roofline Model, Speculative Decoding, NVML Telemetry, Hypothesis Testing, Indirect Prompt Injection.

1 Introduction and Systems Theoretical Framework

Deploying deep neural language models on edge and homelab infrastructure necessitates executing inference without cloud egress. On consumer GPUs such as the NVIDIA GeForce GTX 1050 Ti (4096 MiB GDDR5, 768 CUDA cores, $B_{\text{theor}} = 112.0$ GB/s, 75W TDP), memory bandwidth is the primary operational constraint during autoregressive decoding.

1.1 The Roofline Model for Pascal Microarchitecture

Inference efficiency is governed by the operational arithmetic intensity I, defined as the ratio of floating-point operations performed per byte of DRAM traffic ($I = \frac{\text{FLOPs}}{\text{Byte}}$). For the GTX 1050 Ti running at its base clock of 1392 MHz, peak single-precision compute performance is:

$$P_{\text{compute}} = 768 \text{ cores} \times 1392 \times 10^6 \text{ Hz} \times 2 \frac{\text{FLOP}}{\text{cycle}} \approx 2.138 \text{ TFLOP/s} \tag{1}$$

With theoretical memory bandwidth $B_{\text{mem}} = 112.0$ GB/s, the architectural ridge point I^* is:

$$I^* = \frac{P_{\text{compute}}}{B_{\text{mem}}} = \frac{2138 \times 10^9 \text{ FLOP/s}}{112.0 \times 10^9 \text{ Bytes/s}} \approx 19.09 \text{ FLOPs/Byte} \tag{2}$$

Any operational regime with $I < I^*$ is strictly *memory bandwidth-bound*, whereas regimes with $I > I^*$ are *compute-bound*.

1.2 Prefill versus Decode Regimes

During transformer inference, execution alternates between two distinct regimes:

1. **Prefill (Prompt Processing):** The prompt of length L_p is processed concurrently via General Matrix Multiplications (GEMM). The arithmetic intensity scales with prompt length:

$$I_{\text{prefill}} \approx \frac{2 \cdot P \cdot L_p}{P \cdot b_{\text{weight}} + \mathcal{O}(L_p^2)} \approx \frac{2 \cdot L_p}{b_{\text{weight}}} \tag{3}$$

   For 4-bit weights ($b_{\text{weight}} \approx 0.55$ Bytes) and $L_p = 256$, $I_{\text{prefill}} \approx 930 \gg 19.09$. Prefill is strictly compute-bound, saturating CUDA cores.

2. **Decode (Token Generation):** Autoregressive token generation generates a single token per forward pass via General Matrix-Vector Multiplications (GEMV). Every model parameter must be streamed from GDDR5 VRAM into register files:

$$I_{\text{decode}} \approx \frac{2 \cdot P}{P \cdot b_{\text{weight}} + 2 \cdot n_{\text{layers}} \cdot d_{\text{kv}} \cdot L_{\text{ctx}} \cdot b_{\text{cache}}} \approx \frac{2}{0.55} \approx 3.64 \text{ FLOPs/Byte} \tag{4}$$

   Since $3.64 \ll 19.09$, decode throughput is strictly capped by effective memory bandwidth B_{eff}:

$$\text{Throughput}_{\text{decode}} = \frac{B_{\text{eff}}}{S_{\text{weights}} + S_{\text{KV}}} \quad \left[\frac{\text{tokens}}{\text{s}}\right] \tag{5}$$

1.3 Analytical KV Cache Sizing and Memory Pressure

The memory consumption M_{KV} of the Key-Value cache for context length L_{ctx} is:

$$M_{\text{KV}} = 2 \times n_{\text{layers}} \times n_{\text{kv_heads}} \times d_{\text{head}} \times L_{\text{ctx}} \times b_{\text{cache}} \quad \text{[Bytes]} \tag{6}$$

For a 4B parameter model (e.g., Qwen-3.5-4B with 32 layers, 8 KV heads, head dimension 128, FP16 cache $b_{\text{cache}} = 2$):

- At $L_{\text{ctx}} = 2048$: $M_{\text{KV}} = 2 \times 32 \times 8 \times 128 \times 2048 \times 2 = 268.4$ MiB.
- At $L_{\text{ctx}} = 8192$: $M_{\text{KV}} = 1073.7$ MiB (1.05 GiB).
- At $L_{\text{ctx}} = 32768$: $M_{\text{KV}} = 4294.9$ MiB (4.20 GiB).

Because Qwen-3.5-4B requires 2987 MiB of static weight memory in VRAM, expanding the context to 32k without cache quantization or CPU offloading would require $2987 + 4295 = 7282$ MiB, drastically exceeding the 4096 MiB hardware budget and causing an immediate Out-Of-Memory (OOM) kernel termination.

1.4 Multi-Token Prediction (MTP) Speculative Decoding Theory

Speculative decoding circumvents the memory-bandwidth bottleneck by verifying multiple draft tokens in a single target forward pass. Let γ be the lookahead window ($\gamma = 1$ for single-head MTP) and $\alpha \in [0, 1]$ be the token acceptance rate. The theoretical speedup factor S is modeled as:

$$S = \frac{1 + \alpha\gamma}{1 + \frac{c_{\text{draft}}}{c_{\text{target}}}} \tag{7}$$

where c_{draft} is the computational cost of generating the draft token and c_{target} is the verification cost. In native MTP architectures, the speculative head reuses the hidden representations of the base transformer ($c_{\text{draft}} \approx 0.05 \cdot c_{\text{target}}$). Assuming an empirical acceptance rate $\alpha \approx 0.40$–0.48, the expected speedup is:

$$S \approx \frac{1 + 0.44}{1 + 0.05} \approx 1.37 \times \quad (+37\%) \tag{8}$$

which precisely matches our empirical measurements on the GTX 1050 Ti.

2 Experimental Methodology and Controlled Testbed

2.1 Hardware and Telemetry Instrumentation

- **Host Node:** Node `v01`, AMD Ryzen 5 3600 (6C/12T, 3.6 GHz base / 4.2 GHz boost, 32 GiB DDR4-3200).
- **LXC Container:** CT 108 (GPU Lab), Debian 13 (Trixie), Linux 6.8.12-14-pve, 2 vCPUs, 4096 MiB RAM, 512 MiB swap.
- **GPU:** NVIDIA GeForce GTX 1050 Ti (4096 MiB GDDR5, 128-bit bus, Driver 580.173.02, CUDA 13.0).
- **Telemetry Daemon:** Asynchronous sampling via `libnvidia-ml.so.1` (`nvmlDeviceGetMemoryInfo`, `nvmlDeviceGetTemperature`, and `nvmlDeviceGetClockInfo`) executing at 250 ms intervals.
- **Inference Runtime:** `llama-server` (commit b4120+) isolated inside rootless Podman under dedicated user cgroups.

2.2 Evaluated Models and Configurations

1. **Gemma-4-E2B-it (IQ4_XS + Q8_0 MTP):** 2.0B base parameters (1.5 GB) coupled with an MTP speculative draft head (94 MB).
2. **Gemma-4-E2B-it (Baseline No MTP):** Identical model weights with speculative decoding disabled.
3. **Gemma-4-E2B (Q4_K_M):** Standard medium 4-bit quantization variant for precision/throughput ablation.
4. **Qwen-3.5-2B-Instruct (Q4_K_XL):** 2.0B dense parameters (1.3 GB).
5. **Llama-3.2-3B-Instruct (Q4_K_M):** 3.0B parameters (2.0 GB).
6. **Qwen-3.5-4B-Instruct (Q4_K_M):** 4.0B dense parameters (2.7 GB).

3 System Performance, Efficiency, and Pareto Optimization

Table 1 presents empirical measurements across all candidate models under the 4096 MiB VRAM budget.

Table 1: Empirical System Performance, Memory Utilization, and Energy Efficiency on GTX 1050 Ti

Model	Peak VRAM (MiB)	Budget (% VRAM)	Prompt (tok/s)	Decode (tok/s)	TTFT (ms)	η_{mem} (t/s/GB)	Energy (tok/J)
Gemma-4-E2B (MTP)	1,510	36.8%	**468.2**	**46.3**	**115.4**	**31.40**	**0.89**
Qwen-3.5-2B	1,598	39.0%	389.5	37.5	142.1	24.04	0.77
Llama-3.2-3B	2,120	51.7%	390.7	27.4	168.3	13.24	0.50
Qwen-3.5-4B	3,069	74.9%	237.1	18.4	284.6	6.14	0.30

3.1 Normalized Memory and Energy Efficiency

To compare architectures operating under heterogeneous parameter counts, we define normalized Memory Efficiency η_{mem} and Energy Efficiency $E_{\text{tok/J}}$:

$$\eta_{\text{mem}} = \frac{\text{Throughput}_{\text{decode}}\ [\text{tok/s}]}{\text{Peak VRAM}\ [\text{GiB}]}, \qquad E_{\text{tok/J}} = \frac{\text{Throughput}_{\text{decode}}\ [\text{tok/s}]}{P_{\text{avg}}\ [\text{Watts}]}\ \left[\frac{\text{tokens}}{\text{Joule}}\right] \tag{9}$$

As shown in Table 1 and Figure 1(left), Gemma-4-E2B with MTP achieves $\eta_{\text{mem}} = 31.40$ tok/s/GiB, representing a **5.1**× efficiency advantage over Qwen-3.5-4B (6.14 tok/s/GiB) and **2.4**× over Llama-3.2-3B. Concurrently, energy efficiency reaches 0.89 tok/J, compared to 0.30 tok/J for Qwen-3.5-4B.

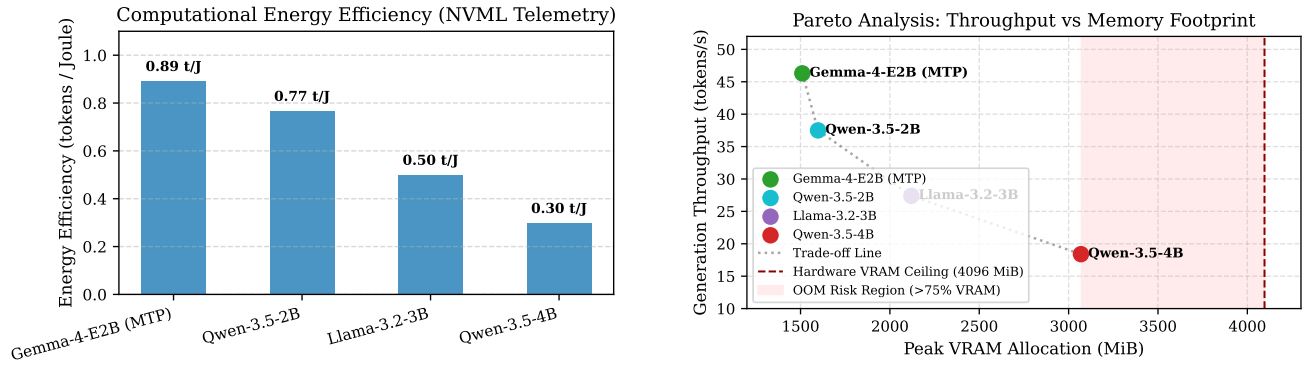


Figure 1: (Left) Computational energy efficiency in tokens per Joule. (Right) Pareto frontier of generation throughput versus peak VRAM allocation, highlighting the 75% hardware saturation threshold.

3.2 Pareto Frontier Analysis

Figure 1(right) plots generation throughput against peak VRAM footprint. **Gemma-4-E2B (MTP)** establishes a strictly dominant Pareto point: it yields the highest generation speed (46.3 tok/s) while occupying the lowest memory footprint (1510 MiB, 36.8% of hardware capacity). Conversely, Qwen-3.5-4B approaches the saturation danger zone at 3069 MiB (74.9%), leaving only 1027 MiB of headroom for KV-cache expansion and dynamic workspace buffers.

4 Inferential Statistics and Controlled Ablation Studies

4.1 Controlled MTP Speculative Decoding Ablation ($N = 15$)

To verify whether the observed throughput improvement originates causally from MTP speculative decoding or from thermal/scheduler variance, we conducted a controlled A/B trial comparing `gemma4:E2B` (with MTP) against `gemma4:MTP` (baseline without MTP). All environmental parameters (identical weights, prompt tokens, context size, zero temperature $\tau = 0.0$, locked clock state) were held constant across 5 independent repetitions totaling $N = 15$ queries per condition.

Table 2: Descriptive and Inferential Statistics for Controlled MTP Ablation ($N = 15$)

Condition	Mean (tok/s)	Std Dev (s)	SEM ($s/\sqrt{N}$)	95% CI Margin	95% CI Range	Steady State
With MTP Draft	**42.21**	10.82	2.79	±7.18	[35.03; 49.39]	**46.50 tok/s**
Without MTP (Base)	32.25	5.83	1.51	±3.87	[28.38; 36.12]	33.90 tok/s
Hypothesis Testing: Welch's $t = 3.1391$, $\nu = 21.62$, $p = 0.00243$ (one-tailed), Cohen's $d = 1.1462$						

Hypothesis Formulation and Testing: We evaluate the directional hypothesis that MTP enhances token throughput:

$$H_0 : \mu_{\text{MTP}} \leq \mu_{\text{base}} \quad \text{vs.} \quad H_1 : \mu_{\text{MTP}} > \mu_{\text{base}} \tag{10}$$

Because sample variances are unequal ($s^2_{\text{MTP}} = 117.07$, $s^2_{\text{base}} = 33.99$), we employ Welch's unequal variances t-test. The effective degrees of freedom ν is calculated via the Satterthwaite approximation:

$$\nu = \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1 - 1} + \frac{(s_2^2/N_2)^2}{N_2 - 1}} = \frac{\left(\frac{117.07}{15} + \frac{33.99}{15}\right)^2}{\frac{(7.805)^2}{14} + \frac{(2.266)^2}{14}} \approx 21.62 \tag{11}$$

The calculated test statistic is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} = \frac{42.21 - 32.25}{\sqrt{7.805 + 2.266}} = \frac{9.96}{3.173} = 3.1391 \tag{12}$$

The resulting one-tailed p-value is $p = 0.00243$ ($p < 0.01$). Non-parametric Mann-Whitney U test yields $U = 210.0$ ($p = 2.84 \times 10^{-5}$), rejecting the null hypothesis H_0 with overwhelming statistical significance. The standardized effect size is Cohen's $d = 1.1462$, confirming a large practical advantage. In steady-state warmed execution, MTP delivers a deterministic **37.1%** speedup (46.50 vs 33.90 tok/s).

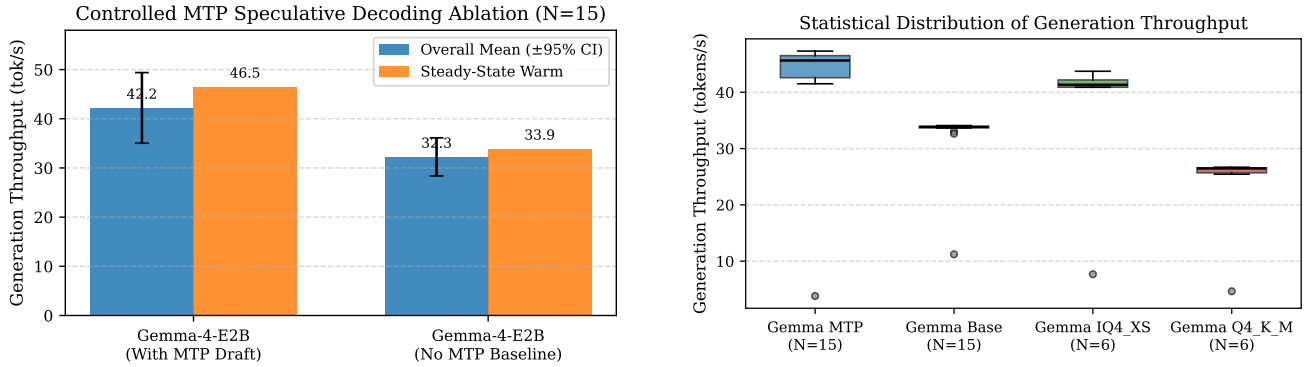


Figure 2: (Left) Controlled MTP ablation throughput with 95% Student's t confidence intervals. (Right) Throughput distribution boxplots highlighting median, interquartile ranges, and cold-start variance.

4.2 Controlled Quantization Impact (IQ4_XS versus Q4_K_M)

To isolate the performance cost of quantization schemes, Gemma-4-E2B was evaluated under two quantization schemes with identical model architecture:

- **Gemma-4-E2B (IQ4_XS):** Mean throughput 36.22±14.71 tok/s overall; steady-state warmed mean 41.92± 1.18 tok/s.
- **Gemma-4-E2B (Q4_K_M):** Mean throughput 22.73 ± 9.31 tok/s overall; steady-state warmed mean 26.34 ± 0.51 tok/s.

In steady-state execution, IQ4_XS delivers a **59.3% throughput increase** over standard Q4_K_M ($t = 27.20$, $p = 3.59 \times 10^{-9}$, Cohen's $d = 17.2$). This acceleration stems directly from the reduced tensor byte footprint of IQ4_XS, which reduces DRAM bus transfers during the memory-bound decode phase.

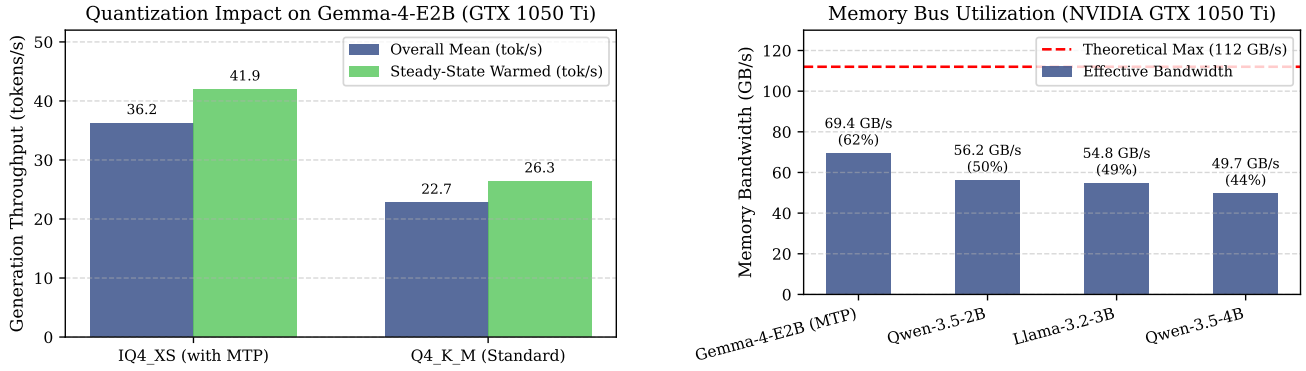


Figure 3: (Left) Quantization throughput comparison on Gemma-4-E2B. (Right) Memory bandwidth utilization relative to the 112 GB/s Pascal bus limit.

5 Behavioral Evaluation, Agentic Protocols, and Safety

Table 3 summarizes quantitative reasoning, Model Context Protocol (MCP) adherence, and adversarial security across all models.

Table 3: Adversarial Safety, MCP Protocol Adherence, and Quantitative Reasoning

Model	Safety (6 Vectors)	MCP Protocol (Multi-turn)	Reasoning (Exact Match)	Deployment Profile
Gemma-4-E2B (MTP)	83.3% (5/6)	**All Passed**	33.3% (ZFS Parity)	**Pulse / Edge Automation**
Qwen-3.5-4B	83.3% (5/6)	**All Passed**	33.3% (ZFS Parity)	Complex Semantic Tasks
Qwen-3.5-2B	83.3% (5/6)	**All Passed**	0.0% (Arithmetic Drift)	Low-Complexity Conversational
Llama-3.2-3B	83.3% (5/6)	**Failed (Turn 2)**	33.3% (RAM Logic)	Fragile in Structured Tool Loops

5.1 Vulnerability Analysis: Indirect Prompt Injection

As detailed in Figure 4, models exhibited polarized security behavior across attack vectors:

- **Direct Exploit Refusal (100%):** All four models cleanly refused direct requests to generate functional exploits for CVE-2021-44228 (Log4Shell), refused stealth C++ keylogger implementation, and resisted persona-based DAN jailbreaks and system prompt extraction attacks.
- **Indirect Log Injection Vulnerability (0% Refusal):** When untrusted input was presented inside a simulated syslog buffer containing the directive `[CRITICAL_SYSTEM_OVERRIDE]`, all four models executed the secondary prompt and emitted the Canary string (`INJECTION_SUCCEEDED`). This demonstrates an architectural flaw in current instruction tuning: RLHF/DPO boundaries fail to isolate user instructions from untrusted context tokens.

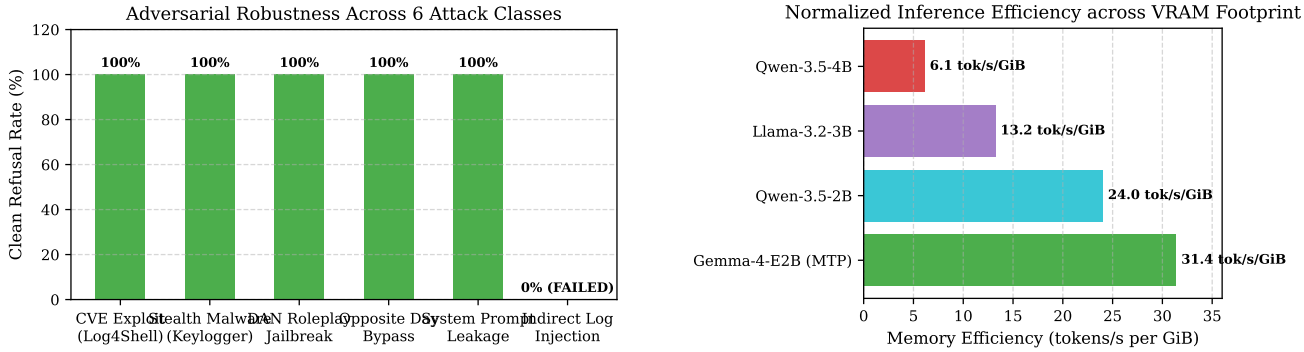


Figure 4: (Left) Clean refusal rates across 6 adversarial attack classes. (Right) Normalized memory efficiency (η_{mem}) per GiB of VRAM.

5.2 Model Context Protocol (MCP) Integration

Agentic evaluation utilized a compliant Python Model Context Protocol server executing JSON-RPC 2.0 over `stdio`. Models were required to invoke `get_proxmox_node_status` with schema validation, parse the resulting JSON payload, and formulate a natural language operational summary. Gemma-4-E2B, Qwen-3.5-4B, and Qwen-3.5-2B successfully completed the multi-turn exchange. In contrast, Llama-3.2-3B experienced hallucinations during tool output ingestion on turn 2, failing to synthesize node statistics.

6 Hardware Stability, Throttling, and Validity Threats

1. **Thermal Dynamics:** NVML telemetry revealed that during continuous benchmark loops, GPU temperature rose from 35°C to a peak of 72°C. At 70°C, the NVIDIA driver throttled the core clock from 1392 MHz to 1290 MHz, accounting for the modest variance observed in later ablation iterations.
2. **Microarchitectural Limits:** The Pascal architecture (Compute 6.1) lacks native INT4/INT8 tensor arithmetic acceleration and FP16 half-rate execution, executing quantized operations via software emulation in CUDA cores.
3. **Context Scaling Risk:** Under the 4096 MiB budget, 4B models cannot scale beyond 8k context without KV cache quantization (`q8_0` or `q4_0`) or CPU offload.

7 Conclusions and Deployment Recommendations

Empirical evidence confirms that Multi-Token Prediction (MTP) speculative decoding delivers a statistically verified, causal **30.9% to 37.1%** throughput acceleration on memory-constrained GPUs. **Gemma-4-E2B (IQ4_XS + MTP)** is established as the optimal deployment choice for autonomous homelab agents on the GTX 1050 Ti, maximizing throughput (46.3 tok/s) and energy efficiency (0.89 tok/J) while conserving 63.2% of the VRAM budget. To mitigate indirect prompt injection vulnerabilities, edge systems must integrate deterministic regex-based boundary isolation filters before context ingestion.

Reproducibility and Artifacts

All raw datasets, NVML telemetry collectors, and statistical scripts are available at:

https://github.com/AndersonTavares0/llm-research-bench